PHYDSC401T — Mathematical Physics–III

Unit 3: Complex Analysis

7th Semester, Assam University (FYUG) — Haflong Government College

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Notes 2 of 20 — written from published university lecture notes (Indian universities and colleges first, then IIT / NPTEL, then IGNOU open-learning material). Full list of sources at the end of these notes.

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Unit syllabus (as in the syllabus file)

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2. Euler's formula, De Moivre's theorem, roots of complex numbers.
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4. Cauchy integral theorem for simply and multiply connected regions, Cauchy integral formula.
5. Taylor and Laurent series, poles, residue theorem, evaluation of integrals.

Top 2 high-mark topics in this unit

1. Cauchy–Riemann equations and Cauchy integral formula / Cauchy integral theorem — AU sets 6-mark Section B questions proving the Cauchy–Riemann equations and showing $\sin z$ is analytic, and stating and proving Cauchy's integral formula.
2. Residue theorem, residues at poles, and evaluation of integrals — AU sets several 6-mark Section B questions on residues at simple and double poles, on the sum of residues being zero, and on residue evaluation of real integrals from 0 to 2π.

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1. Complex numbers: modulus, argument, conjugate, graphical picture

A complex number is $z = x + iy$, where x and y are real numbers and $i^2 = -1$. Here x is the real part, written $\operatorname{Re} z = x$, and y is the imaginary part, $\operatorname{Im} z = y$. Two complex numbers are equal exactly when their real parts are equal and their imaginary parts are equal.

Graphical representation. Plot $z = x + iy$ as the point (x, y) in the Argand plane, with the real axis horizontal and the imaginary axis vertical. Addition of complex numbers is then addition of vectors in the plane, and the distance of the point from the origin is the modulus.

Modulus. The modulus, or absolute value, of $z = x + iy$ is

$$|z| = \sqrt{x^2 + y^2} = \sqrt{z\bar{z}}$$

where $\bar{z} = x - iy$ is the complex conjugate of z. The modulus is the distance from the origin, so $|z| \geq 0$, and $|z| = 0$ only for $z = 0$. It obeys $|z_1 z_2| = |z_1|\,|z_2|$ and the triangle inequality $|z_1 + z_2| \leq |z_1| + |z_2|$.

Argument. The argument of $z \neq 0$ is the angle θ that the vector z makes with the positive real axis, so that $x = r\cos\theta$ and $y = r\sin\theta$ with $r = |z|$. It is written $\arg z = \theta$. The argument is only fixed up to multiples of 2π; the value in $-\pi < \theta \leq \pi$ is called the principal argument, $\operatorname{Arg} z$. In terms of x and y, $\tan\theta = y/x$, with the quadrant decided by the signs of x and y.

Conjugate. The conjugate $\bar{z} = x - iy$ is the reflection of z in the real axis. It satisfies $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$, $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$, $\bar{\bar{z}} = z$, and $z + \bar{z} = 2x = 2\operatorname{Re} z$, $z - \bar{z} = 2iy = 2i \operatorname{Im} z$.

$$z + \bar{z} = 2\operatorname{Re} z, \qquad z\bar{z} = x^2 + y^2 = |z|^2$$

So the sum and the product of a complex number and its conjugate are both real: the sum is twice the real part, and the product is the square of the modulus. (AU 2022, Sec A.) The same idea gives the division rule: to divide by $z_2 \neq 0$, multiply top and bottom by $\bar{z}_2$, so that $z_1/z_2 = z_1\bar{z}_2/|z_2|^2$.

Example. The square root of $-4 - 3i$. Let $\sqrt{-4 - 3i} = a + ib$ with a, b real. Squaring, $(a + ib)^2 = a^2 - b^2 + 2iab = -4 - 3i$, so $a^2 - b^2 = -4$ and $2ab = -3$. Also $a^2 + b^2 = |-4 - 3i| = 5$. Adding and subtracting, $2a^2 = 1$ and $2b^2 = 9$, so $a = \pm 1/\sqrt{2}$ and $b = \mp 3/\sqrt{2}$, with opposite signs because $ab < 0$. Hence

$$\sqrt{-4 - 3i} = \pm\frac{1 - 3i}{\sqrt{2}}$$

Checking one of them: $((1 - 3i)/\sqrt{2})^2 = (1 - 6i - 9)/2 = -4 - 3i$, as required. (AU 2022, Sec B, 6 marks.)

2. Euler's formula and De Moivre's theorem, roots of complex numbers

Euler's formula. For real θ,

$$e^{i\theta} = \cos\theta + i\sin\theta$$

so every complex number can be written in polar form $z = re^{i\theta}$ with $r = |z|$ and $\theta = \arg z$. Multiplication is then easy: $r_1 e^{i\theta_1} \cdot r_2 e^{i\theta_2} = r_1 r_2 e^{i(\theta_1+\theta_2)}$: moduli multiply and arguments add. Division is the same with arguments subtracted. Euler's formula also gives $\cos\theta = (e^{i\theta} + e^{-i\theta})/2$ and $\sin\theta = (e^{i\theta} - e^{-i\theta})/(2i)$, which is how the trigonometric functions of a complex variable are defined.

De Moivre's theorem. For a positive integral index n,

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$$

To prove it, use induction on n. It is true for $n = 1$. Suppose it is true for $n = k$. Then $(\cos\theta + i\sin\theta)^{k+1} = (\cos k\theta + i\sin k\theta)(\cos\theta + i\sin\theta)$, and multiplying out gives $(\cos k\theta\cos\theta - \sin k\theta\sin\theta) + i(\sin k\theta\cos\theta + \cos k\theta\sin\theta) = \cos(k+1)\theta + i\sin(k+1)\theta$ by the addition formulae for sine and cosine. So it holds for $k+1$, and hence for every positive integer n. (AU 2023, Sec B, 4 marks; AU 2022, Sec A.) In exponential form this is just $(e^{i\theta})^n = e^{in\theta}$, and the theorem extends to negative integers by taking reciprocals and to rational powers by taking roots.

Roots of complex numbers. The nth roots of $z = re^{i\theta}$ are $r^{1/n}e^{i(\theta+2k\pi)/n}$ for $k = 0, 1, \ldots, n-1$: n distinct roots, equally spaced on a circle of radius $r^{1/n}$. In particular the nth roots of unity are $e^{2k\pi i/n}$, and their sum is zero for $n > 1$.

Trigonometric functions of a complex variable. Writing $z = x + iy$ and using Euler's formula, $\sin z = (e^{iz} - e^{-iz})/(2i)$ and $\cos z = (e^{iz} + e^{-iz})/2$. Expanding $e^{iz} = e^{ix-y} = e^{-y}(\cos x + i\sin x)$ and $e^{-iz} = e^{y}(\cos x - i\sin x)$ gives, after separating real and imaginary parts,

$$\sin(x+iy) = \sin x\cosh y + i\cos x\sinh y$$

$$\cos(x+iy) = \cos x\cosh y - i\sin x\sinh y$$

where $\cosh y = (e^y + e^{-y})/2$ and $\sinh y = (e^y - e^{-y})/2$. So $\sin z$ and $\cos z$ are unbounded for complex z, unlike their real versions: for example $\sin(iy) = i\sinh y$ grows without bound as y grows. (AU 2023, Sec A.)

Example. If $2\cos\theta = x + 1/x$ and $2\cos\phi = y + 1/y$, then $x^p y^q + 1/(x^p y^q) = 2\cos(p\theta + q\phi)$. The equation $2\cos\theta = x + 1/x$ is $x^2 - 2x\cos\theta + 1 = 0$, whose roots are $x = \cos\theta \pm i\sin\theta = e^{\pm i\theta}$. Taking $x = e^{i\theta}$ and $y = e^{i\phi}$, we get $x^p y^q = e^{i(p\theta+q\phi)}$ and $1/(x^p y^q) = e^{-i(p\theta+q\phi)}$, and their sum is $2\cos(p\theta + q\phi)$ by Euler's formula. (AU 2022, Sec B, 6 marks.)

3. Functions of a complex variable: limits, continuity, derivatives

A function of a complex variable is a rule $w = f(z)$ assigning to each z in some region a complex number $w = u + iv$. Writing $z = x + iy$, the function is fixed by its two real parts, $u = u(x, y)$ and $v = v(x, y)$, so that $f(z) = u(x, y) + iv(x, y)$. For example $f(z) = z^2$ is $u = x^2 - y^2$, $v = 2xy$.

Limit and continuity. We say $f(z) \to L$ as $z \to z_0$ if $f(z)$ can be made as close as we like to L by taking z close enough to z_0, from any direction in the plane. That last phrase is the whole difference from real calculus: z can approach z_0 along any path, and the limit must be the same along all of them. A function is continuous at z_0 if $\lim_{z \to z_0} f(z) = f(z_0)$. Sums, products and quotients (with non-zero denominator) of continuous functions are continuous, and polynomials are continuous everywhere.

Derivative. The derivative is defined exactly as in real calculus,

$$f'(z_0) = \lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$$

but again $\Delta z \to 0$ from any direction. This is a strong demand: a function can have partial derivatives u_x, u_y, v_x, v_y and still fail to be differentiable as a complex function, because the limit along the real direction and along the imaginary direction may disagree. The condition that makes them agree is the Cauchy–Riemann equations, which is the subject of the next section. When $f'(z)$ exists at every point of a region, f is said to be analytic, or holomorphic, in that region.

A warning example. $f(z) = \bar{z} = x - iy$ is continuous everywhere, but it is differentiable nowhere as a complex function: approaching along the real direction gives $f'(z) = 1$, while approaching along the imaginary direction gives $f'(z) = -1$. So $|z|^2 = z\bar{z}$ is likewise not differentiable as a complex function, even though it is a perfectly smooth function of x and y. Complex differentiability is much stricter than real differentiability.

4. Analytic functions and the Cauchy–Riemann equations (with proof)

Analytic function. A function $f(z) = u + iv$ is analytic at a point z_0 if it is differentiable not only at z_0 but at every point of some neighbourhood of z_0. It is analytic in a region if it is analytic at every point of the region. A function analytic in the whole complex plane is called entire: polynomials are entire, e^z, $\sin z$ and $\cos z$ are entire, and $1/z$ is analytic everywhere except at $z = 0$. A point where a function fails to be analytic is a singularity of the function.

Cauchy–Riemann equations. Let $f(z) = u(x, y) + iv(x, y)$ be differentiable at $z_0 = x_0 + iy_0$, with u and v having continuous first partial derivatives. Then at z_0,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

To prove it, compute the derivative in two ways. First let $\Delta z = \Delta x$ be real, so z approaches z_0 along the real direction. Then $\Delta u + i\Delta v$ divided by Δx tends to $u_x + iv_x$. Next let $\Delta z = i\Delta y$ be purely imaginary, so z approaches along the imaginary direction. Then $(\Delta u + i\Delta v)/(i\Delta y)$ tends to $v_y - iu_y$. Since the derivative exists, these two limits must be the same complex number, so equating real and imaginary parts gives $u_x = v_y$ and $v_x = -u_y$, which is the same pair of equations. The argument also runs the other way: if u and v have continuous first partial derivatives satisfying the Cauchy–Riemann equations in a region, then f is differentiable there, and its derivative is

$$f'(z) = u_x + iv_x = v_y - iu_y$$

So the Cauchy–Riemann equations are the necessary and sufficient condition for complex differentiability, given continuous partial derivatives. They are what the real and imaginary parts of an analytic function must satisfy at every point where the function is analytic. (AU 2022, Sec B, 6 marks.)

Consequences. Differentiating $u_x = v_y$ with respect to x and $u_y = -v_x$ with respect to y, and using equality of mixed partials, gives $u_{xx} + u_{yy} = 0$: the real part of an analytic function is harmonic, and the same holds for v. So u and v are harmonic conjugates of each other. If f is analytic and $|f|$ is constant in a region, then f itself is constant there; and if f is analytic with $f'(z) = 0$ everywhere in a region, then f is constant in that region.

Example: $\sin z$ is analytic. From the previous section, $\sin z = \sin x \cosh y + i \cos x \sinh y$, so $u = \sin x \cosh y$ and $v = \cos x \sinh y$. Then $u_x = \cos x \cosh y$ and $v_y = \cos x \cosh y$, so $u_x = v_y$; and $u_y = \sin x \sinh y$ while $v_x = -\sin x \sinh y$, so $u_y = -v_x$. The Cauchy–Riemann equations hold everywhere, with continuous partial derivatives, so $\sin z$ is analytic in the whole plane, and its derivative is $f'(z) = u_x + iv_x = \cos x \cosh y - i \sin x \sinh y = \cos z$. (AU 2022, Sec B, 6 marks.) The same calculation shows $\cos z$ is analytic with derivative $-\sin z$, and e^z is analytic with derivative e^z.

Example: z^2 is analytic, $\bar{z}$ is not. For $f(z) = z^2$, $u = x^2 - y^2$ and $v = 2xy$, so $u_x = 2x = v_y$ and $u_y = -2y = -v_x$: the equations hold everywhere. For $f(z) = \bar{z}$, $u = x$ and $v = -y$, so $u_x = 1$ but $v_y = -1$: the equations fail everywhere, which is why $\bar{z}$ is differentiable nowhere.

5. Cauchy integral theorem and Cauchy integral formula (with proof)

Contour integrals. Let C be a smooth curve in the plane, parametrised by $z = z(t)$, and let f be continuous on C. The contour integral is $\int_C f(z)\,dz = \int (u\,dx - v\,dy) + i\int (v\,dx + u\,dy)$, computed by substituting the parametrisation. A closed contour is one that returns to its starting point. The direction matters: reversing C changes the sign of the integral.

Cauchy integral theorem. If f is analytic inside and on a simple closed contour C, then

$$\oint_C f(z)\,dz = 0$$

The idea of the proof is Green's theorem in the plane. Writing the integral as the real line integrals above and applying Green's theorem turns each of them into a double integral over the region D enclosed by C: the real part becomes $\iint_D (-v_x - u_y)\,dx\,dy$ and the imaginary part becomes $\iint_D (u_x - v_y)\,dx\,dy$. Both integrands are zero by the Cauchy–Riemann equations, so the whole integral is zero. The theorem says that for an analytic function, the integral between two points does not depend on the path, only on the endpoints, exactly as for a conservative field in vector calculus. It fails if f has a singularity inside C: for example $\oint_{|z|=1} dz/z = 2\pi i$, not zero, because $1/z$ is not analytic at $z = 0$ inside the circle.

Multiply connected regions. The theorem extends to a region with holes. If C is an outer closed contour and $C_1, C_2, \ldots$ are inner closed contours around the holes, all traversed so that the region stays on the left, then $\oint_C f(z)\,dz = \oint_{C_1} f(z)\,dz + \oint_{C_2} f(z)\,dz + \cdots$ for f analytic in the region between them. In words, the integral round the outside equals the sum of the integrals round the holes. This is the form of the theorem the syllabus names, and it is what lets us deform a contour: an integral round a complicated contour can be replaced by the integral round a small circle round each singularity inside it.

Cauchy integral formula. If f is analytic inside and on a simple closed contour C, and a is a point inside C, then

$$f(a) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - a}\,dz$$

This is the most useful formula in the unit: the values of an analytic function inside a region are completely fixed by its values on the boundary. To prove it, cut out a small circle C_ρ of radius ρ round a, inside C. By the multiply connected form of the integral theorem, the integral over C equals the integral over C_ρ. On C_ρ, write $z = a + \rho e^{i\theta}$, so $dz = i\rho e^{i\theta} d\theta$ and $f(z)/(z-a)\,dz = i f(a + \rho e^{i\theta})\,d\theta$. As $\rho \to 0$, $f(a + \rho e^{i\theta}) \to f(a)$ uniformly, because f is continuous, so the integral tends to $i f(a) \int_0^{2\pi} d\theta = 2\pi i f(a)$. Dividing by $2\pi i$ gives the formula. (AU 2022, Sec B, 6 marks.)

Derivatives from the formula. Differentiating under the integral sign gives the formula for derivatives, $f^{(n)}(a) = \dfrac{n!}{2\pi i} \oint_C \dfrac{f(z)}{(z-a)^{n+1}}\,dz$. Two consequences follow at once. First, an analytic function has derivatives of all orders, and those derivatives are themselves analytic. Second, if $|f(z)| \le M$ on a circle of radius R round a, then $|f^{(n)}(a)| \le n! M / R^n$ (Cauchy's estimate). Letting $R \to \infty$ for an entire bounded function gives $f'(a) = 0$ everywhere, which is Liouville's theorem: a bounded entire function is constant. The fundamental theorem of algebra follows from it: a non-constant polynomial must have a zero, or else $1/p(z)$ would be a bounded entire function.

Example. $\oint_{|z|=2} dz/(z^2 - 1)$ is zero. The integrand has simple poles at $z = \pm 1$, both inside the circle, with residues $1/2$ and $-1/2$ (see Section 7), which cancel. Equivalently, partial fractions give $1/(z^2 - 1) = \tfrac{1}{2}[1/(z-1) - 1/(z+1)]$, and each term integrates to $2\pi i$ times its residue by the integral formula, so the total is $2\pi i(\tfrac{1}{2} - \tfrac{1}{2}) = 0$. (AU 2022, Sec B, 6 marks.)

6. Taylor and Laurent series

Taylor series. If f is analytic inside a disc $|z - a| < R$, then f can be written as a power series in that disc,

$$f(z)=\sum_{n=0}^{\infty}a_n(z-a)^n,\qquad a_n=\frac{f^{(n)}(a)}{n!}=\frac{1}{2\pi i}\oint_C\frac{f(w)}{(w-a)^{n+1}}\,dw$$

where C is any circle $|w-a|=\rho<R$ round a. The series converges to $f(z)$ for $|z-a|<R$, and R is the distance from a to the nearest singularity of f. The standard series, valid for all z, are $e^z=\sum z^n/n!$, $\sin z=\sum(-1)^n z^{2n+1}/(2n+1)!$ and $\cos z=\sum(-1)^n z^{2n}/(2n)!$. (AU 2022, Sec A.)

Example: $\cos z$ about $z=\pi/4$. Put $w=z-\pi/4$, so $z=\pi/4+w$. Then $\cos z=\cos(\pi/4+w)=\cos(\pi/4)\cos w-\sin(\pi/4)\sin w=\frac{1}{\sqrt2}(\cos w-\sin w)$. Expanding $\cos w$ and $\sin w$ in their power series,

$$\cos z=\frac{1}{\sqrt2}\left[\sum_{n=0}^{\infty}\frac{(-1)^n w^{2n}}{(2n)!}-\sum_{n=0}^{\infty}\frac{(-1)^n w^{2n+1}}{(2n+1)!}\right],\qquad w=z-\frac{\pi}{4}$$

so $\cos z=\frac{1}{\sqrt2}[1-w-w^2/2+w^3/6+w^4/24-\cdots]$. The series converges for all z, since $\cos z$ is entire. (AU 2022, Sec A.)

Laurent series. If f is analytic in an annulus $r<|z-a|<R$, then f can be written as a series with both positive and negative powers,

$$f(z)=\sum_{n=-\infty}^{\infty}a_n(z-a)^n,\qquad a_n=\frac{1}{2\pi i}\oint_C\frac{f(w)}{(w-a)^{n+1}}\,dw$$

where C is any circle in the annulus. The part with $n\ge0$ is the analytic part, and the part with $n<0$ is the principal part; the principal part is what carries the singularity at a. When f is analytic at a itself, the principal part vanishes and the Laurent series is just the Taylor series. The annulus is decided by the singularities: each region between two consecutive singularities (in distance from a) has its own Laurent expansion, and the expansion changes when a singularity is crossed. (AU 2022, Sec A.)

Example: $f(z)=1/[(z+1)(z+3)]$. Partial fractions give $f(z)=\frac12[1/(z+1)-1/(z+3)]$. The singularities are at $z=-1$ and $z=-3$, so there are two regions about the origin. For $|z|<1$, both terms expand in powers of z: $1/(z+1)=\sum(-1)^n z^n$ and $1/(z+3)=\frac13\sum(-1)^n(z/3)^n$, so

$$f(z)=\frac12\sum_{n=0}^{\infty}(-1)^n\left[1-\frac{1}{3^{n+1}}\right]z^n,\qquad |z|<1$$

For $1<|z|<3$, the $1/(z+1)$ term must expand in powers of $1/z$, because $|1/z|<1$ there, while the $1/(z+3)$ term still expands in powers of z: $1/(z+1)=\sum(-1)^n z^{-n-1}$, so

$$f(z)=\frac12\left[\sum_{n=0}^{\infty}(-1)^n z^{-n-1}-\frac13\sum_{n=0}^{\infty}(-1)^n\left(\frac{z}{3}\right)^n\right],\qquad 1<|z|<3$$

which has both a principal part and an analytic part. (AU 2022, Sec B, 6 marks.)

Example: $f(z)=1/[(z-1)(z-2)]$. Partial fractions give $f(z)=1/(z-2)-1/(z-1)$. For $|z|<1$ this is $\sum[1-2^{-(n+1)}]z^n$; for $1<|z|<2$ it is $-\frac12\sum(z/2)^n-\sum z^{-n-1}$; and for $|z|>2$

it is $\sum (2^n - 1) z^{-n-1}$. The same function has three different Laurent expansions in its three regions, and the region must always be stated with the expansion.

7. Singularities, poles and residues

Singularities. An isolated singularity of f at $z = a$ is a point where f is not analytic but is analytic in some punctured neighbourhood $0 < |z - a| < r$. From the Laurent series $f(z) = \sum a_n (z-a)^n$ about a, there are three kinds. If the principal part is absent, the singularity is removable: f can be redefined at a to become analytic there, as with $\sin z / z$ at $z = 0$. If the principal part has finitely many terms, ending at $(z-a)^{-m}$, the singularity is a pole of order m: a simple pole for $m = 1$, a double pole for $m = 2$. If the principal part has infinitely many terms, the singularity is essential: $e^{1/z}$ at $z = 0$ is the standard example, and near an essential singularity f takes every complex value (with at most one exception) infinitely often. A singularity is a pole when $|f(z)| \to \infty$ as $z \to a$, and an essential singularity when $f(z)$ has no limit at all as $z \to a$. (AU 2022, Sec A.)

Residue. The residue of f at an isolated singularity a is the coefficient a_{-1} of $(z-a)^{-1}$ in its Laurent expansion,

$$\operatorname{Res}(f, a) = a_{-1} = \frac{1}{2\pi i} \oint_C f(z)\, dz$$

where C is a small circle round a. It is the only Laurent coefficient that survives integration round a, because $\oint (z-a)^n dz = 0$ for $n \neq -1$ and $2\pi i$ for $n = -1$.

Finding residues at poles. If f has a simple pole at a, the residue is $\operatorname{Res}(f, a) = \lim_{z \to a} (z-a) f(z)$. If $f = g/h$ with $g(a) \neq 0$, $h(a) = 0$ and $h'(a) \neq 0$, then $\operatorname{Res}(f, a) = g(a)/h'(a)$. If f has a pole of order m at a, then $\operatorname{Res}(f, a) = \dfrac{1}{(m-1)!} \lim_{z \to a} \dfrac{d^{m-1}}{dz^{m-1}} [(z-a)^m f(z)]$. (AU 2022, Sec A.)

Example: residues of $\cot z$. $\cot z = \cos z / \sin z$ has simple poles where $\sin z = 0$, that is at $z = n\pi$ for every integer n. At each pole the residue is $\cos(n\pi) / \cos(n\pi) = 1$, using the g/h' rule with $g = \cos z$ and $h' = \cos z$. So $\operatorname{Res}(\cot z, n\pi) = 1$ at every pole. (AU 2022, Sec A.)

Example: residues of $z^2/[(z-1)(z-2)(z-3)]$. There are simple poles at $1, 2, 3$. The residues are $\operatorname{Res}(f, 1) = 1/[(-1)(-2)] = 1/2$, $\operatorname{Res}(f, 2) = 4/[(1)(-1)] = -4$, and $\operatorname{Res}(f, 3) = 9/[(2)(1)] = 9/2$. Their sum is $1/2 - 4 + 9/2 = 1$. The residue at infinity is defined as minus the sum of all the finite residues, so $\operatorname{Res}(f, \infty) = -1$, and the sum of all residues including infinity is zero. In general, for a rational function the sum of all residues, including the residue at infinity, is always zero. (AU 2022, Sec B, 6 marks.)

Example: residue at a double pole. For $f(z) = z^2/[(z+1)^2(z-2)]$, the pole at $z = -1$ is double. Put $g(z) = z^2/(z-2)$; then the residue is $g'(-1)$. Now $g'(z) = [2z(z-2) - z^2]/(z-2)^2 = (z^2 - 4z)/(z-2)^2$, so $g'(-1) = (1+4)/9 = 5/9$. Hence $\operatorname{Res}(f, -1) = 5/9$. (AU 2022, Sec B, 6 marks.)

Example: residues of $e^{zt}/[z^2(z^2+2z+2)]$. The poles inside $|z|=3$ are a double pole at $z=0$ and simple poles at $z=-1\pm i$. Expanding $e^{zt}/(z^2+2z+2)$ about $z=0$ gives the residue at the double pole as $(t-1)/2$. At $z=-1+i$ the residue is $e^{(-1+i)t}/4$, and at $z=-1-i$ it is $e^{(-1-i)t}/4$, in each case from $e^{zt}/[z^2(2z+2)]$ at the pole. Their sum is $(t-1)/2+e^{-t}\cos t/2$. (AU 2022, Sec B, 6 marks; the integral itself is evaluated in the next section.)

8. Residue theorem and evaluation of real integrals

Cauchy residue theorem. If f is analytic inside and on a simple closed contour C, except for isolated singularities $a_1,\dots,a_n$ inside C, then

$$\oint_C f(z)\,dz = 2\pi i \sum_{k=1}^{n} \operatorname{Res}(f,a_k)$$

The proof is contour deformation: cut out a small circle round each singularity, use the multiply connected form of Cauchy's theorem, and on each small circle only the a_{-1} term of the Laurent series survives. The residue theorem turns a contour integral into arithmetic: find the poles inside, add their residues, multiply by $2\pi i$. (AU 2022, Sec A.)

Real integrals from 0 to 2π. An integral $\int_0^{2\pi} R(\cos\theta,\sin\theta)\,d\theta$ of a rational function of sine and cosine becomes a contour integral round the unit circle by putting $z=e^{i\theta}$, so that $\cos\theta=(z+z^{-1})/2$, $\sin\theta=(z-z^{-1})/(2i)$ and $d\theta=dz/(iz)$. The integral is then $2\pi i$ times the sum of the residues inside $|z|=1$.

Example. $\int_0^{2\pi} d\theta/(5-4\sin\theta)$. With $z=e^{i\theta}$ the integral becomes $\oint_{|z|=1} dz/(-2z^2+5iz+2)$. The denominator vanishes at $z=2i$ and $z=i/2$; only $z=i/2$ lies inside the unit circle, with residue $1/(3i)$. Hence the integral is $2\pi i\cdot 1/(3i)=2\pi/3$. (AU 2022, Sec B, 6 marks.)

Example. Using the residue theorem, $(1/2\pi i)\oint_{|z|=3} e^{zt}\,dz/[z^2(z^2+2z+2)]$ is the sum of the residues found in the previous section, $(t-1)/2+e^{-t}\cos t/2$. So the contour integral itself is $\pi i\,[(t-1)+e^{-t}\cos t]$. (AU 2022, Sec B, 6 marks.)

Integrals over the whole real line. For $\int_{-\infty}^{\infty} f(x)\,dx$ with f rational and decaying fast enough, close the contour with a large semicircle in the upper half-plane. The integral over the semicircle vanishes as its radius tends to infinity, and the integral over the real line equals $2\pi i$ times the sum of the residues in the upper half-plane. If the integrand is even, the integral from 0 to ∞ is half of that.

Example. $\int_0^{\infty} dx/(x^2+a^2)^2=\pi/(4a^3)$ for $a>0$. The integrand $1/(z^2+a^2)^2$ has a double pole at $z=ia$ in the upper half-plane, with residue $1/(4ia^3)$, from differentiating $1/(z+ia)^2$ at $z=ia$. So $\int_{-\infty}^{\infty} dx/(x^2+a^2)^2=2\pi i\cdot 1/(4ia^3)=\pi/(2a^3)$, and halving for the even integrand gives $\pi/(4a^3)$. (Extra, UPSC Mathematics Paper-II, 2018.)

Example. $\oint_{|z|=2}(e^z+1)\,dz/[z(z+1)(z-i)^2]$. The poles inside are a simple pole at $z=0$ with residue -2, a simple pole at $z=-1$ with residue $i(1+e^{-1})/2$, and a double pole at $z=i$ with residue

$g'(i)$, where $g(z) = (e^z + 1)/(z^2 + z)$. The integral is $2\pi i$ times the sum of the three residues. (Extra, UPSC Mathematics Paper-II, 2015.)

A pole is a pole of order m exactly when. An isolated singularity z_0 of f is a pole of order m if and only if $f(z) = \phi(z)/(z - z_0)^m$ near z_0, where ϕ is analytic and non-zero at z_0. In one direction this is the definition of a pole from the Laurent series, taking $\phi(z) = (z - z_0)^m f(z)$; in the other direction, if f has that form then its Laurent series about z_0 starts at $(z - z_0)^{-m}$ with non-zero coefficient, so the pole is of order m. (Extra, UPSC Mathematics Paper-II, 2019.)

9. Worked AU questions, in full

Every AU question in the blueprint for this unit is answered in the sections above. This section gathers them in one place, so the unit can be revised question by question. Sources are [H] PHSHCC-401T, 4th Sem, 2022, CBCS, and [B] MATDSC-101T, 1st Sem, 2023, FYUG.

Short questions (Section A, 2 marks each).

- **Define modulus and argument of a complex number. [H]** $|z| = \sqrt{x^2 + y^2}$ is the distance from the origin in the Argand plane; $\arg z = \theta$ is the angle the vector z makes with the positive real axis, so $z = |z|e^{i\theta}$. See Section 1.
- **Show that the sum and product of a complex number and its conjugate are both real. [H]** $z + \bar{z} = 2\,\mathrm{Re}\,z$ and $z\bar{z} = |z|^2$. See Section 1.
- **State and explain De Moivre's theorem. [H]** $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$ for positive integral n, proved by induction. See Section 2.
- **What is singularity of an analytic function? Define pole. [H]** A singularity is a point where the function is not analytic but is analytic in a punctured neighbourhood. A pole of order m is a singularity whose Laurent principal part ends at $(z - a)^{-m}$; $|f(z)| \to \infty$ as $z \to a$. See Section 7.
- **State Taylor and Laurent expansions. [H]** Taylor: $f(z) = \sum a_n (z - a)^n$ with $a_n = f^{(n)}(a)/n!$, valid where f is analytic. Laurent: $f(z) = \sum_{-\infty}^{\infty} a_n (z - a)^n$, valid in an annulus where f is analytic, with a principal part carrying the singularity. See Section 6.
- **Expand $\cos z$ in a Taylor series about $z = \pi/4$. [H]** $\cos z = \frac{1}{\sqrt{2}}[1 - w - w^2/2 + w^3/6 + \cdots]$ with $w = z - \pi/4$. See Section 6.
- **State Cauchy residue theorem. [H]** $\oint_C f(z)\,dz = 2\pi i$ times the sum of the residues inside C. See Section 8.
- **How will you find the residue at a simple pole? [H]** $\mathrm{Res}(f, a) = \lim_{z\to a}(z - a)f(z)$, or $g(a)/h'(a)$ when $f = g/h$. See Section 7.
- **Find the residue at each pole of $f(z) = \cot z$. [H]** Simple poles at $z = n\pi$, residue 1 at each. See Section 7.
- **Express $\sin(x + iy)$ in the form $A + iB$. [B]** $A = \sin x \cosh y$, $B = \cos x \sinh y$. See Section 2.

Long questions (Section B, 4–6 marks each).

- **State De Moivre's theorem for positive integral index and prove it. [B, 4 marks]** See Section 2, induction proof.
- **If $2\cos\theta = x + 1/x$ and $2\cos\phi = y + 1/y$, prove $x^p y^q + 1/(x^p y^q) = 2\cos(p\theta + q\phi)$; and find the square root of $-4 - 3i$. [H, 6 marks]** $x = e^{\pm i\theta}$, $y = e^{\pm i\phi}$; and $\sqrt{-4-3i} = \pm(1 - 3i)/\sqrt{2}$. See Sections 1 and 2.
- **Show that the real and imaginary parts of an analytic function satisfy the Cauchy–Riemann equations; and show that $\sin z$ is analytic. [H, 6 marks]** Derivative along the real and imaginary directions must agree, giving $u_x = v_y$ and $u_y = -v_x$; for $\sin z$ both equations hold with $f'(z) = \cos z$. See Section 4.
- **State and prove Cauchy's integral formula. [H, 6 marks]** $f(a) = \frac{1}{2\pi i}\oint_C f(z)/(z-a)\,dz$; proof by cutting out a small circle round a. See Section 5.
- **Evaluate $\int_C dz/(z^2-1)$, C the circle $x^2 + y^2 = 4$; and expand $f(z) = 1/[(z+1)(z+3)]$ as a Laurent series valid for (i) $|z| < 1$ and (ii) $1 < |z| < 3$. [H, 6 marks]** The integral is 0, residues $1/2$ and $-1/2$ cancelling; the two Laurent expansions are in Section 6.
- **Evaluate the residues of $z^2/[(z-1)(z-2)(z-3)]$ at $z = 1, 2, 3$ and infinity and show their sum is zero; and find the residue of $f(z) = z^2/[(z+1)^2(z-2)]$ at its double pole. [H, 6 marks]** Residues $1/2, -4, 9/2$, sum 1, residue at infinity -1; double-pole residue $5/9$. See Section 7.
- **Using the residue theorem, calculate $(1/2\pi i)\int_C e^{zt}\,dz/[z^2(z^2+2z+2)]$, C the circle $|z| = 3$; and using residue calculus, evaluate $\int_0^{2\pi} d\theta/(5 - 4\sin\theta)$. [H, 6 marks]** The first is $(t-1)/2 + e^{-t}\cos t/2$; the second is $2\pi/3$. See Sections 7 and 8.

Extra important questions (non-AU). $\int_0^\infty dx/(x^2+a^2)^2 = \pi/(4a^3)$ (UPSC 2018); $\oint_{|z|=2}(e^z+1)\,dz/[z(z+1)(z-i)^2]$ (UPSC 2015); a pole of order m exactly when $f(z) = \phi(z)/(z-z_0)^m$ with ϕ analytic and non-zero at z_0 (UPSC 2019); Taylor and Laurent expansions of $1/[(z-1)(z-2)]$ in $|z| < 1$, $1 < |z| < 2$ and $|z| > 2$. All four are worked in Sections 6 and 8.

Note on the [H] paper. The same 2022 paper also sets Laplace transform questions. Laplace transforms are not in the FYUG PHYDSC401T syllabus for this unit, so they are not covered in these notes.

10. Revision formula box

$z = x + iy$, $|z| = \sqrt{x^2 + y^2}$, $\bar{z} = x - iy$, $z\bar{z} = |z|^2$, $z = re^{i\theta}$.
Euler: $e^{i\theta} = \cos\theta + i\sin\theta$. De Moivre: $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$.
$\sin(x + iy) = \sin x\cosh y + i\cos x\sinh y$.
Cauchy–Riemann: $u_x = v_y$, $u_y = -v_x$. Then $f'(z) = u_x + iv_x$.
Cauchy integral theorem: $\oint_C f(z)\,dz = 0$ for f analytic inside and on C.
Cauchy integral formula: $f(a) = \frac{1}{2\pi i}\oint_C f(z)/(z-a)\,dz$.
Taylor: $f(z) = \sum f^{(n)}(a)(z-a)^n/n!$. Laurent: $f(z) = \sum_{-\infty}^{\infty} a_n(z-a)^n$.
Residue at a simple pole: $\lim_{z\to a}(z-a)f(z)$; at a pole of order m: $\frac{1}{(m-1)!}\lim_{z\to a}\frac{d^{m-1}}{dz^{m-1}}[(z-a)^m f(z)]$.
Residue theorem: $\oint_C f(z)\,dz = 2\pi i\sum \mathrm{Res}(f, a_k)$.
$\int_0^{2\pi} d\theta/(5 - 4\sin\theta) = 2\pi/3$. $\int_0^{\infty} dx/(x^2 + a^2)^2 = \pi/(4a^3)$.

Sources

Written from published university lecture notes, in the order the user set on 2026-11-05: Indian colleges and universities first, then IIT and NPTEL, then IGNOU, then MIT OCW only for the written form of the current step.

- Assam University question papers in the AU bank: PHSHCC-401T, 4th Sem, 2022, CBCS [H]; MATDSC-101T, 1st Sem, 2023, FYUG [B]; PHYDSC-101T, 1st Sem, 2023, FYUG [A]. All AU questions in these notes are from these papers, as listed in the blueprint.
- Syllabus: PHYDSC401T Mathematical Physics–III, Unit 3, Assam University FYUG, in ~/workspace/user/files/7th_semester_syllabus.md. Reference books named in the syllabus: M. R. Spiegel, Complex Variables (McGraw Hill); G. B. Arfken, Mathematical Methods for Physicists (Academic Press).
- Standard complex analysis texts used for the derivations: Spiegel, Complex Variables; Arfken and Weber, Mathematical Methods for Physicists; Kreyszig, Advanced Engineering Mathematics (complex analysis chapters); Brown and Churchill, Complex Variables and Applications.
- Extra questions: UPSC Mathematics Paper-II, 2015, 2018 and 2019, as listed in the blueprint.
- Method: top 2 high-mark topics first (Cauchy–Riemann and Cauchy integral formula; residue theorem and evaluation of integrals), then Assam University exact questions, then extra important questions, per the blueprint built 2026-11-05.